ATI Allvac

2020 Ashcraft Avenue,
Monroe, NC 28111-5030
US

CERTIFICATE OF TEST

Cert No.10628 Rev.1

Certification Auditor : Todd E. Langer\es
Date : November 25, 2013

Batch - 16625
Heat - 13JE

Customer Name & Address	Purchase Order No	Sales Order No	Sales Order Line No
Allegheny Technologies Limited Granby Avenue Garretts Green Birmingham B33 OSP GB	10729	12463	3.2

Size (mm)	Cross section	No Pcs	Weight (kgs)	Alloy
25.400	Rectangle	2	*1001*	ATI 6-4™ Alloy

Specifications

Spec Name	Rev	Class	Compliance Condition
ABS 5125	6	MATERIAL=A	Compliant
AIMS 03-18-001	5		Compliant
AMS 2242	G		Compliant
AMS 2631	D	CLASS=A1	Compliant
AMS 4911	M		Compliant
AMS-STD-2154	A	TYPE=I, CLASS=AA	Compliant
AMS-T-9046	B	CODE=AB-1	Compliant
AS 9100	C		Compliant
ASTM B265	2011	GRADE=5	Compliant
ASTM F1472	2008*1		Compliant
AUK/SA/001-3	09/00		Compliant
BAE/AG/QC/SC1 PARTS 1-7	3		Compliant
BS 2TA56	2009		Compliant
EN 10204	10-2004		Compliant
ETTC 4	2		Compliant
ISO 9001	11/15/08		Compliant
LN 9297	91 AMD A1		Compliant
NF L 00-015	C		Compliant
SABRE2	1		Compliant
SQA 50	12		Compliant
WL 3.7164 TEIL 1	11/90		Compliant

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ATI 26

Remarks:

AMS-T-9046 is a cancelled specification replaced by current AMS specifications.
AMS-T-9046 was a word for word translation of MIL-T-9046 that was cancelled on 9/17/1999.

Approved Certificate
AIRBUS Approval 30213
Certified that the whole of the supplies detailed hereon have been manufactured, inspected, and tested and unless otherwise stated above conform in all respects to the specifications, drawings, and contract/order relevant thereto.

This material was produced under BAE Systems (Air Systems) approved quality management system per ISO 9001/AS 9100.

Material certified in accordance with EN 10204 Type 3.1.

EN 10204 certification does not indicate Pressure Equipment Directive approval for this material.

As Shipped Condition

Heat Treat	Heat Treat Cycles	Hot Work Type
Vacuum Creep Flattening	Heat to 774C, Hold 6hrs min., Vacuum Cool	Rolled

Remarks:

Final Heat Treatment and Final Inspection performed at ATI Precision Finishing (Monaca, PA)

Surface Finish

Mechanically Descaled

Remarks:

Mechanical descale performed at ATI Precision Finishing (Rochester, PA and/or Monaca, PA).

Identity Details

Material Identity

A4A - 25.400 MM X 1347.79 MM X 3113.09 MM

A4B - 25.400 MM X 1346.20 MM X 3282.95 MM

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Melt Method Details

Primary Melt	Facility	Address
Vacuum Arc Remelt	ATI Albany Operations	530 34th Avenue, SW, Albany, OR 97322 US
Remelt	**Facility**	**Address**
Vacuum Arc Remelt	ATI Albany Operations	530 34th Avenue, SW, Albany, OR 97322 US

CHEMISTRY

Piece ID		TOP	BOTTOM
Sample Source		INGOT	INGOT
Test Facility		ATI - WAHCHANG	ATI - WAHCHANG
Elements	UOM		
Al	%	6.14	6.24
V	%	4.08	3.99
Fe	%	0.17	0.15
O	%	0.18	0.19
C	%	0.02	0.02
N	%	<0.01	<0.01
Y	%	<0.001	<0.001
Ti	%	BAL	BAL

Test Comments: Metallics are analyzed by ICP, gases are analyzed by inert gas fusion or combustion.

Remarks:

Other Elements (Each) = 0.10% maximum
Other Elements (Total) = 0.40% maximum

Test Methods: C = ASTM E1941 (2010); O/S = ASTM E1409 (2008); H = ASTM E1447 (2009); Plasma Emission = ASTM E2371 (2013), ASTM E1097(2012); XRF = ASTM E539 (2011); ICP = ASTM E2594 (2009)

Piece ID			A4 - TOP	A4 - BOTTOM
Sample Source			PLATE	PLATE
Test Facility			DIRATS	DIRATS
Elements	UOM	Method		
H	%	GAS	0.0059	0.0061

Remarks:

H = ASTM E1447 (2009)

MECHANICAL

Plant: ,,,1400-1450°C (760-788C), 6 HRS. MIN., VCF

Tensile

Piece ID	Sample Direction	Sample Location	Test Temperature	Ultimate Strength (MPa)	.2% Yield Strength (MPa)	4D-Elongation (%)	5D-Elongation (%)	Reduction of Area (%)	Test Facility
A4 - TOP	L	C	ROOM	995	916	17	15	33	DIRATS
A4 - TOP	LT	C	ROOM	1036	976	15	14	24	DIRATS
A4 - MID	LT	C	ROOM	1036	974	17	15	26	DIRATS
A4 - BOT	L	C	ROOM	998	923	18	16	36	DIRATS
A4 - BOT	LT	C	ROOM	1015	958	17	15	26	DIRATS

Remarks:

L = Longitudinal
LT = Longitudinal Transverse
C = Center
ASTM E8/E8M (2011)

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Fracture Toughness

Piece ID	Sample Direction	Test Temperature	K1C (MPa)	Valid/Invalid (None)	Test Facility
A4	T - L	ROOM	54	VALID	DIRATS

METALLOGRAPHY

Plant: ,,,1400-1450°C (760-788C), 6 HRS. MIN., VCF

STR TI-PREN 3114-3

Sample Direction	Sample Location	Predominant (Figure Number)	Pass/Fail/Accept (None)	Test Facility
A4	L	3T10, 3T12, 3T24, 3T32	PASS	DIRATS

continued...

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Sample Direction	Sample Location	Predominant (Figure Number)	Pass/Fail/Accept (None)	Test Facility
A4	T	3T10, 3T11, 3T12, 3T13, 3T32	PASS	DIRATS

Remarks:

L = Longitudinal

T = Transverse

Microstructure was evaluated in accordance with prEN3114-3, Rev. P1 and found acceptable to the requirements of AIMS 03-18-001.

Grain Size

Piece ID	Sample Direction	Uniform ASTM (ASTM Number)	Test Facility
A4	L	9.0	DIRATS
A4	T	9.0	DIRATS

Test Description: Average Grain Size

Remarks:

L = Longitudinal

T = Transverse

ASTM E112 (2012), ASTM E930 (1999/2007), ASTM E1181 (2002/2008)

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Alpha Case

Remarks:

L = Longitudinal

S-S = Surface to Surface

Material evaluated and found to be free of alpha case.

Remarks:

Beta Transus Calculated by Chemistry:
992 C. = TOP
999 C. = BOTTOM

Material evauated in accordance with ETTC4 and found acceptable and most similar to Photomicrographs P10, P12, P13, P21, P29, P30.

Ultrasonic

Test Facility: ATI Precision Finishing – Monaca, PA

Remarks:

Material ultrasonically inspected in accordance with AMS 2631 Class A1 and found acceptable.

Material ultrasonically inspected in accordance with AMS-STD-2154 Type I Class AA and found acceptable.

Outside Source Addresses

Outside Testing Laboratory

Facility	Address
ATI Wah Chang	1600 NE Old Salem Rd., Albany, OR 97321 USA
Dirats & Co., Inc.	41 Airport Rd., Westfield, MA 1086 USA

Subcontracted Operations

Facility	Address
ATI Albany Operations	530 34th Avenue SW, Albany, OR 97321 USA
ATI Allegheny Ludlum - Washington	500 Green St., Washington, PA 15301 USA
ATI Precision Finishing	499 Delaware Ave., Rochester, PA 15074 USA
ATI Precision Finishing	288 West New Castle St., Zelienople, PA 16063 USA
ATI Precision Finishing	2070 Pennsylvania Ave., Monaca, PA 15061 USA

Other Attached Files

Attached File Description(s):

1) NF L 00-015C

Special Remarks

Material was rolled at ATI Allegheny Ludlum, located at 500 Green St., Washington, PA 15301.

Remarks

Material has been produced, sampled, inspected and tested in accordance with the customer purchase order and referenced specifications and conforms to the requirements unless otherwise noted in this Certificate of Test.

If customer purchase order does not specifically reference a revision to a specification, ATI Allvac will work to the latest revision on file and in effect at the time of order placement.

Any chemical elements analyzed and found to have values below the actual limits of detection may be reported as < less than or reported at the detection level.

When values are reported to the significant places called for in the specifications, rounding will be done in accordance with ASTM E29.

This is to certify that during the manufacturing, handling, testing and inspection, this material did not come in direct contact with mercury or any device employing a single boundary of containment.

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Remarks

ATI Allvac products have not come in contact with radioactive materials during manufacturing or processing.

No weld repair has been performed on this material.

Material melted and manufactured in the United States of America unless otherwise noted in this Certificate of Test.

The recording of false, fictitious, or fraudulent statements or entries on this document may violate Federal statutes, including but not limited to Title 18, Chapter 47 of the United States Code and may be punishable as a felony.

This Certificate of Test shall not be reproduced, except in full, without the written approval of ATI Allvac Quality.

Material Safety Data Sheets (MSDS) - View or print from our site: www.atimetals.com
Printed copies are available upon request from the ATI Allvac Sales Department.

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PUBLISHED BY THE AEROSPACE STANDARDIZATION OFFICE - Technopolis 54 - 199, rue Jean-Jaques Rousseau - 92138 ISSY -LES -MOULINEAUX CEDEX - TEL. : 01-47-65-70-00- FAX 01-48-82-08-40

Supplier: ATI Allvac **Establishment:** 2020 Ashcraft Avenue PO Box 5030 Monroe, NC 28112	**Declaration Number:** 55015 **Number of Sheets:** 5
Customer: Allegheny Technologies Ltd **Establishment:** Keys House, Granby Avenue Garretts Green Birmingham GB	**DECLARATION OF CONFORMITY** (NF L 00-015C)
Contract Number: 10460	**No. and Date of Delivery Note:** N/A

Description	Reference Number or Type	Serial or Lot No.	QUANTITY	COMMENTS
Titanium Plate	6Al - 4V Alloy	16625	2 PLATES 1001 KG	

We declare that the cited supply is in comformity with the requirements of the contract and that, after verification and tests, it meets in every respect the specified requirements and the applicable standards and regulations subject to exceptions, reservations or concessions listed in the present declaration of conformity.

Supplier's Quality Officer

Name and Title: Justin Smith
Quality Assurance
Date: July 20, 2012

Signature

Reserved for the Authorized inspection agency

Name and Title:
Date

Signature: